# Abby Koch, EIT

Medical Devices: Quality, R&D & Clinical Engineering | Engineer (EIT) & Clinical Prosthetics & Orthotics
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## Profile

Engineer (EIT) with a clinical Post-graduate Diploma in Prosthetics and Orthotics. Ran 8 custom prosthetic projects at once with clinics across North America, co-led an ethics-approved user trial, and managed an AS9100 calibration program in aerospace manufacturing.

## Key Skills

**Quality & Regulatory:** AS9100 calibration program, SOPs and quality procedures, electronic change control, root cause analysis, mock internal audits, Health Canada Class I determination, ISO 13485 familiarity.
**Clinical:** Socket fabrication, casting and modification, carbon fibre lamination, alignment and fitting, gait analysis, outcome measures, patient training, interdisciplinary teams.
**Design & Research:** Fusion 360 parametric design, 3D scanning and printing, F-Scan pressure mapping, ethics applications (TCPS 2: CORE), RStudio, OPRC platform presenter.
**Other:** EIT (P.Eng. in progress), Aviation Human Factors training, native English, basic French, Canadian citizen.

## Work Experience

**Victoria Hand Project** Victoria, BC
Mechanical Engineer (EIT) May 2024 – Jul 2025 (Part-time from Sep 2024)

- Ran 8 custom prosthetic projects at once with fitting partners, from design to fitting, training each clinician and passing their feedback to engineering.
- Made a check socket for each of the 8 patients, then designed the final sockets in Fusion 360 with the project's parametric macro.
- Partnered with the CEO on clinical adoption across North America, building relationships with 30+ prosthetic clinics and converting 8 into fitting partners.
- Built an onboarding tracker and database replacing scattered records, following each clinic from first contact to project start and logging post-fitting issues.
- Traced a remote fitting issue to short cabling; added buffer cabling to every shipment and a pre-fitting readiness call that became standard practice.
- Determined that Health Canada Class I licensing was not required for the organization's devices, documenting the regulatory rationale.

**Custom Prosthetic Services** Victoria, BC
Clinical Assistant Jun 2025 – Jul 2025

- Cast, modified and laminated a carbon fibre transtibial socket under a certified prosthetist, picking the foot and setting up the alignment.
- Scheduled the patient's visits so the socket was fitted before my contract ended; welcomed patients and took intake information.
- Evaluated 3 scanning systems against clinic workflow needs; recommended the system the clinic purchased and supported its initial workflow integration.

**Magellan Aerospace Ltd.** Winnipeg, MB
Process Engineering Specialist (EIT) | Continual Improvement Facilitator Jan 2022 – Dec 2023
Engineering Co-op Student May 2021 – Dec 2021

- Calibrated the Instron and other test equipment at 100% AS9100 compliance, ran mechanical tests, and trained technicians on test methods.
- Wrote test procedures for metallurgical, hardness and mechanical testing; completed acceptance testing that cleared a $1.2M laser system purchase.
- Wrote SOPs and quality procedures and routed each change through formal electronic change control, incorporating reviewer feedback before release.
- Facilitated root cause analysis with operators, quality and engineering (fishbone, 5 Whys, Pareto), tracking corrective actions to closure.

## Education

**British Columbia Institute of Technology** Burnaby, BC

Post-graduate Diploma in Clinical Prosthetics and Orthotics (With Distinction) Sep 2024 – May 2026

- Fabricated sockets for 3 transtibial and 2 transfemoral patient projects.
- Winnipeg Prosthetics & Orthotics practicum: fabricated 2 transtibial sockets and a partial foot prosthesis.
- Practicums at Winnipeg P&O and Queen Alexandra Centre (Island Health): intake, assessment, fitting, gait analysis, outcome measures and patient training; booked patients' visits myself.
- Capstone: Co-led an ethics-approved user trial of 2 prototypes (pressure mapping, comfort ratings); wrote the ethics application; co-presented at OPRC, Montreal (Aug 2026).
- Post-polio KAFO: gait analysis, hand casting and load-relief modifications reduced knee hyperextension from 17° to 3° in midstance.

**University of Manitoba** Winnipeg, MB

B.Sc. in Biosystems Engineering (Dean's Honour List) 2018 – 2022

- Coursework: biomechanics, physiology and human factors. Undergraduate Teaching Assistant, Design in Engineering.